

# From Legal PDFs to Verifiable Answers

Evaluating answer quality, page-level grounding and latency across 100 legal questions

## HYPOTHESIS Can a page-traceable Legal RAG pipeline balance answer quality, source grounding, validity and latency?

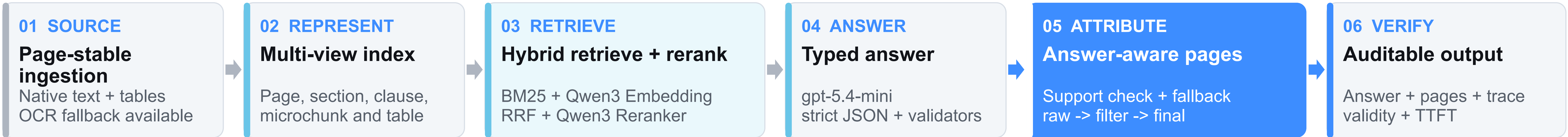
H2: Does answer-aware attribution sharpen citations while retaining at least one correct page per annotated question?

### FINDING

H1 is partly supported: grounding missed its 0.85 threshold. H2 holds for precision, compactness and 95/95 hit coverage.

## METHODOLOGY

100 questions | 30 publicly available legal PDFs | 590 pages | 95 with page annotations



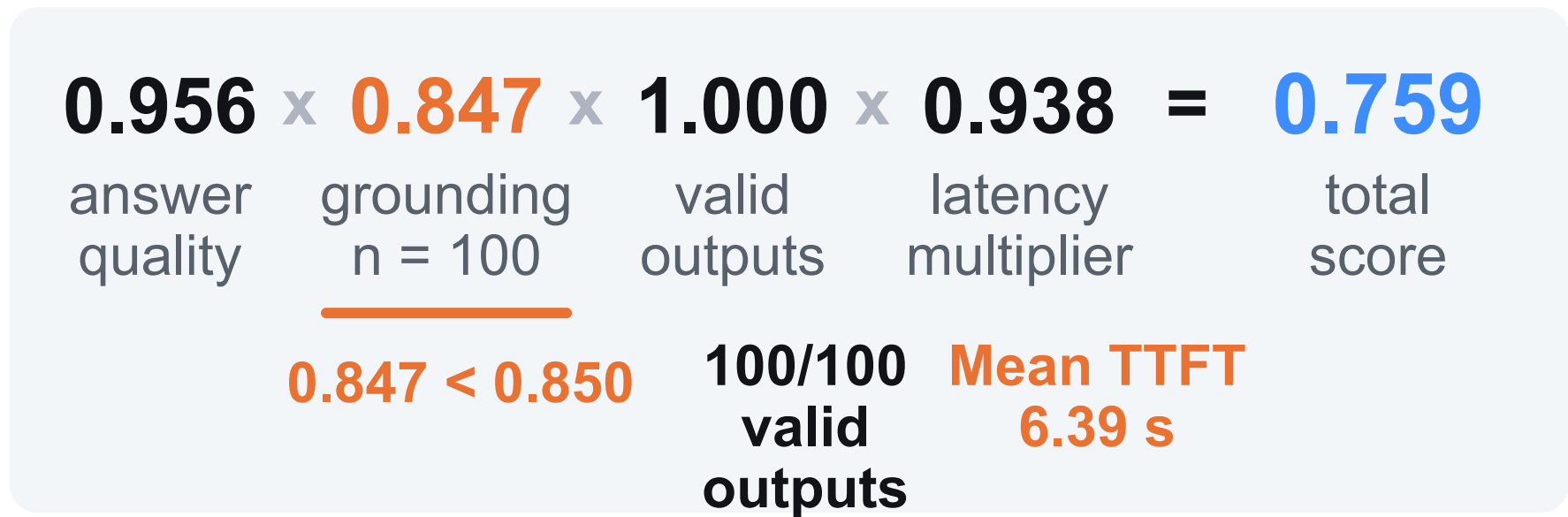
CONTINUOUS PROVENANCE | document\_id + page\_number

The same identifiers survive every stage, so a final citation can be traced back to the exact PDF page.

Evaluation A: end-to-end profile on 100 questions. Evaluation B: paired raw -> filter -> final page sets on the same 95 annotated questions.

### RESULTS | SYSTEM

Grounding limits an otherwise strong system



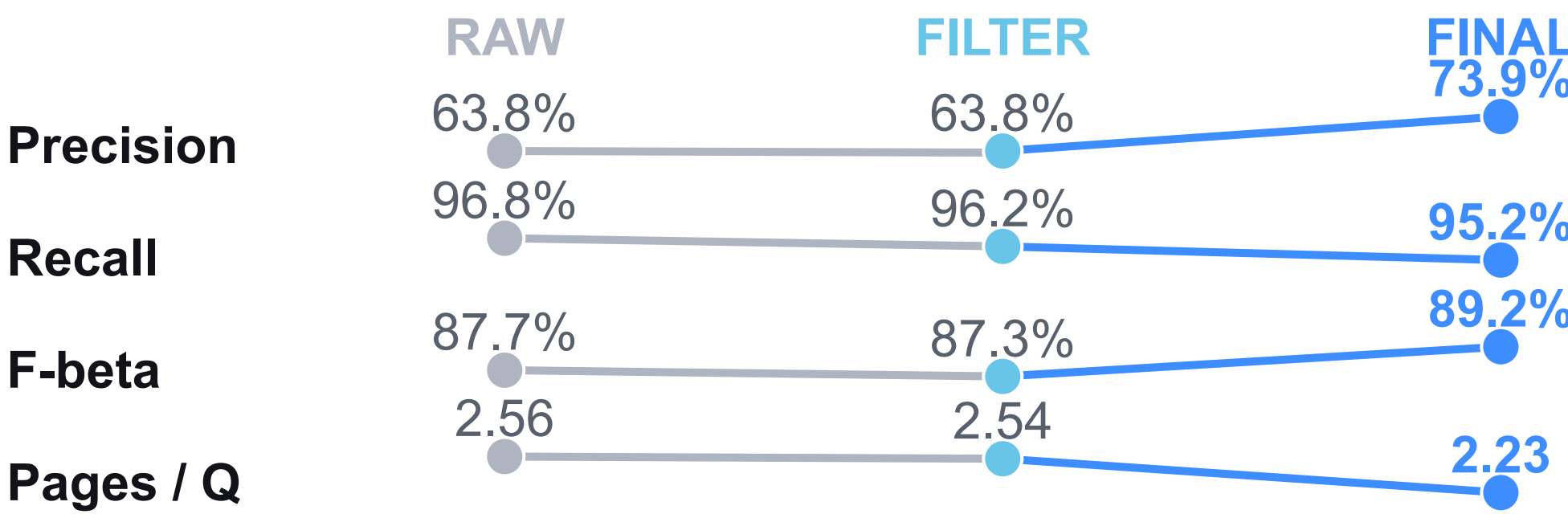
0.943 deterministic | 0.987 free text

H1 is partly supported: quality, validity and latency pass; benchmark grounding misses 0.85.

### RESULTS | ATTRIBUTION

Answer-aware selection removes citation noise

Macro averages | n = 95 annotated questions

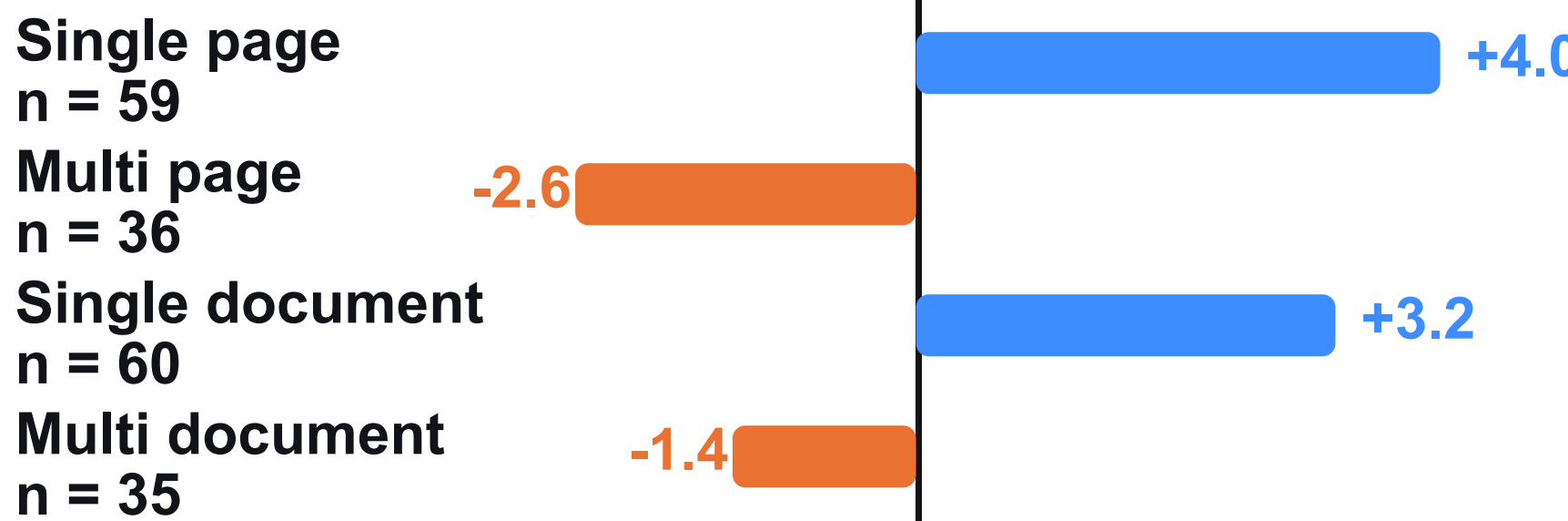


+10.2 pp precision 95% CI +6.3 to +14.4 | -0.33 pages / question 95% CI -0.46 to -0.20 | 95/95 hit coverage retained F-beta gain uncertain

### RESULTS | BOUNDARY

Narrowing helps local - and hurts distributed evidence

Exploratory F-beta delta (pp), raw -> final



Multi-page regression is the clearest failure boundary.

Do not conflate: benchmark grounding 0.847 = all 100 answers; macro page F-beta 0.892 = final attribution on 95 annotated questions.

### DECISION

Retrieve broadly. Attribute explicitly. Narrow conservatively when evidence spans pages or documents.

### LIMITATIONS

#### ONE SAVED RUN

Retrospective thresholds and questions; no repeated seeds.

#### NO CAUSAL ABLATION

Pipeline stages were not independently compared against a baseline.

#### LIMITED GENERALITY

TTFT only; OCR not exercised; one DIFC corpus and one language setting.

[1] Lewis et al., NeurIPS 2020 [2] Karpukhin et al., EMNLP 2020 [3] Cormack et al., SIGIR 2009 [4] Gao et al., EMNLP 2023

[5] Pipitone & Alami, LegalBench-RAG 2024 [6] Zhang et al., Qwen3 Embedding 2025 | Full references in appendix.